South Okanagan Bat Hibernacula Project 2019-2025

2025-26 Update

Mandy Kellner Camila Hurtado

2026-10-28

Table of contents

1 Land Acknowledgement **3**

2 Background **3**

3 Methodology **4**
3.1 Acoustic Monitoring 4
3.2 Summary of Results 6
3.2.1 Data Submission 6

4 Results **6**

5 Discussion **8**
5.1 Recommendations 8

6 Acknowledgements **8**

7 Appendix A **9**

Literature Cited **31**

1 Land Acknowledgement

Biodiversity Pathways respectfully acknowledges that our work takes place on Treaty 8 and Douglas Treaties Territories as well as the traditional and unceded territories of First Nations and Métis Peoples across all regions of British Columbia, whose histories, languages, and cultures are deeply connected to the biodiversity we monitor. We acknowledge the traditional teachings of the lands that we work on, and that reciprocal, meaningful, and respectful relationships with Indigenous peoples make our work possible. We are deeply grateful for their stewardship of these lands, and we are committed to supporting Indigenous-led monitoring programs, while learning Indigenous ways of knowing, being, and doing.

2 Background

Growing evidence indicates that most BC bat species overwinter within the province, with several remaining occasionally active through the winter. This activity has been documented acoustically in both cave and non-cave habitats (Lausen and Rae 2018), at cliff roosts suspected to be hibernacula in the Peace region (WLRS unpublished data), in coastal forests (WLRS unpublished data), and at five parks in southern BC (BC Parks unpublished data). Further year-round monitoring efforts likely exist beyond those noted here.

Although this activity spans diverse habitats (cliffs, old forests, and waterbodies) little is known about the specific non-cave microsites western bats use in winter (Nagorsen et al. 1994; Weller et al. 2018). Documented non-cave roosts include talus and tree roosts in southeast Alaska (Blejwas et al. 2021), tree cavities in southeastern BC (deFreitas 2023), a possible talus slope in the same region (Lausen and Hill 2021), and anthropogenic sites such as the siding and trim of houses on the coast (BC Community Bat Program unpublished data). Apart from several large hibernacula in southern BC mines, few hibernacula have been identified, suggesting that most of the region's bats hibernate singly or in small, scattered groups. This mirrors findings from the western US, where wintering Myotis were found singly in 45% of records or in small groups with a median of two, and Townsend's big-eared bats in slightly larger groups with a median of three (Weller et al. 2018) .

Understanding how bats use winter habitat has practical value. It can inform the management of recreational and industrial activities, and clarifying how bats use non-cave features such as rock crevices, talus, and trees may improve understanding of how *Pseudogymnoascus destructans* (the fungus that causes white-nose syndrome) spreads in BC, as well as potential treatment options.

Three studies in the Okanagan region, spanning the winter of 2011/12 to the present, form the basis of this report. The first, at Skaha Bluffs in 2011/12, revealed extensive winter activity, recording 1,551 bat passes over 105 sample nights (Sarell 2012) . At least eleven species were detected between 7 December 2011 and 31 March 2012 and are therefore thought to hibernate in the area.

BC Parks followed with a multi-year acoustic monitoring study across the Okanagan, running from 2019 through the winter of 2023 to locate potential hibernacula. It recorded ten species in late fall and early winter, and intermittently through the winter, all in rocky outcrops and canyons; this activity was initially interpreted as evidence of hibernacula near the detectors (Paterson 2024) . Field surveys in 2024/25, however, pointed to an alternative: bats may have been moving rapidly through these sites, using them as flight corridors rather than roosting nearby. Long-term detector data may therefore be a weaker indicator of local hibernation than first believed, and confirming whether

bats actually roost in these features, and pinpointing the microsites they use, would substantially advance understanding of their winter ecology.

This motivated a shift in approach. In 2024/25, Skurikhina and Kellner (2025) established a capture site at Doctor's Wall to locate hibernacula microhabitats directly. The pilot assessed the feasibility of fall and winter capture and tracking in South Okanagan parks and protected areas, and used radiotelemetry to identify the microsites tagged bats used. Specifically, it aimed to:

- assess whether bats could be captured in late fall and early winter in the South Okanagan, particularly Skaha Park and Haynes Lease Ecological Reserve;
- begin identifying the microsites bats use in late fall and winter; and
- test the assumption underlying acoustic monitoring, that late fall and early winter detections indicate bats hibernating nearby, since relocating tagged bats by telemetry would confirm use of the immediate area.

In 2025/26 we also invested in data capture and sharing. All acoustic data from long-term monitoring sites (Paterson 2024; Skurikhina and Kellner 2025) were submitted to the North American Bat Monitoring Program database and to WildTrax, supporting larger-scale analyses and allowing the files to be stored and re-analysed in future. Data from Sarell (2012) were not added to these databases, but both of the other projects datasets were submitted to the BC Wildlife Species Inventory database.

Finally, we developed a data visualization tool that future results can be added to. Though simple at present, it is designed to be readily expanded into an interactive resource for examining patterns in winter bat detections across Okanagan parks and protected areas.

3 Methodology

3.1 Acoustic Monitoring

Passive acoustic monitoring was conducted using Wildlife Acoustic Song Meter MiniBat units, placed out for one to multiple nights, to confirm bat presence and aid in selecting suitable netting microsites. Occasional active monitoring was done during mist-netting, using a handheld Wildlife Acoustic Echometer Touch Pro to monitor bat activity at each capture site. Monitoring effort was not recorded for active monitoring (Table 1). For the Sarell (2012) project acoustic monitoring was done with SM2 Bat and SM2Bat+ units from wildlife acoustics. Most sites were sampled as full-spectrum wave files but the Fortress, Wave, No Climbs and Great White site sin 2012 were sampled only with zero crossing.

Table 1: Parks/ Protected Areas, winter acoustic monitoring sites, and years monitored. LT = Long-term monitoring (duration of months); ST = short-term monitoring (duration of days to weeks).

Site	Latitude	Longitude	Years monitored							Data type	Project report/data source (BC WSI project ID 6430)
			2011-12	2018-19	2019-20	2020-21	2021-22	2022-23	2024-25		
Tplaqin / Enderby Cliffs Park											
Enderby Cliffs	50.5900	-119.0833			x					LT	Paterson 2024[1]
Kalamalka Lake park											
Cougar Canyon	50.1806	-119.2751		x		x	x			LT	Paterson 2024[1]
Okanagan Mountain Park											
Wildhorse Canyon	49.7450	-119.6838			x					LT	Paterson 2024[1]
Okanagan Mountain	49.6741	-119.6092		x		x	x			LT	Paterson 2024[1]
Haynes Lease Ecological Reserve											
Haynes Lease 11	49.0940	-119.5226		x						LT	Paterson 2024[1]
Haynes Lease 21	49.0923	-119.5202			x					LT	Paterson 2024[1]
Haynes Lease 31	49.0930	-119.5179				x	x	x		LT	Paterson 2024[1]
Skaha Bluffs Park											
Fern Gully	49.4501	-119.5617	x							ST	Sarrell 2012[2]
Fortress	49.4371	-119.5676	x							ST	Sarrell 2012[2]
Wave	49.4385	-119.5653	x							ST	Sarrell 2012[2]
No Climbs	49.4390	-119.5627	x							ST	Sarrell 2012[2]
Great White	49.4415	-119.5620	x							ST	Sarrell 2012[2]
Whitewash	49.4348	-119.5666	x							ST	Sarrell 2012[2]
Unknown	49.4357	-119.5627	x							ST	Sarrell 2012[2]
Blazing	49.4403	-119.5631	x							ST	Sarrell 2012[2]
Go Anywhere	49.4434	-119.5620	x							ST	Sarrell 2012[2]
Cave Hill	49.4491	-119.5612	x							ST	Sarrell 2012[2]
Skaha - Great White	49.4424	-119.5619			x					LT	Paterson 2024[1]
Skaha - Fern Gully / Doctors Wall	49.4434	-119.5656		x		x	x	x		LT	Paterson 2024[1]
Gillies Creek Road Detector	49.4301	-119.5622							x	ST	Skurikhina and Kellner 2025[3]
Gillies Creek Open Forest Detector	49.4312	-119.5619							x	ST	Skurikhina and Kellner 2025[3]
Smythe Drive Spring Detector	49.4284	-119.5668							x	ST	Skurikhina and Kellner 2025[3]

[1]Paterson. 2024. Okanagan winter bat monitoring: 5 year summary 2018-2023. Prepared for Kirk Safford, BC Parks.
[2]Sarell. 2012. Late Winter Activity of Bats At Skaha Bluffs. Prepared for: BC Parks Penticton, BC
[3]Skurikhina and Kellner. 2025. South Okanagan Bat Hibernacula Microsite Pilot Project. 2024-25 field update.

Acoustic results for the sites sampled in 2012 were analyzed using Sonobat (version 3.05 Great Basin) if they were full spectrum wave files, and if they were zero-cross files thery were manually

reviewd using Analook software. For all other projects data were analyzed using Kaleidoscope Pro (Wildlife Acoustics, version 5.6.8) with bat auto ID Bats of North America 5.8.1 modified for the region. The species list included Pallid Bats (*Antrozous pallidus*, ANPA), Townsend's Big-eared Bat (*Corynorhinus townsendii*, COTO), Big Brown Bat (*Eptesicus fuscus*, EPFU), Hoary Bat (*Lasiurus cinereus*, LACI), Silver-haired Bat (*Lasionycteris noctivagans*, LANO), Californian Myotis (*Myotis californicus*, MYCA), Western small-footed Myotis (*M. ciliolabrum*, MYCI), Long-eared Myotis (*M. evotis*, MYEV), Little Brown Myotis (*M. lucifugus*, MYLU), Fringed Myotis (*M. thysanodes*, MYTH), Long-legged Myotis (*M. volans*, MYVO), and Yuma Myotis (*M. yumanensis*, MYYU). All files were manually reviewed by Brian Paterson and Mandy Kellner.

3.2 Summary of Results

We summarized results differently depending on the data available for each study. For Skurikhina and Kellner (2025) and Paterson (2024) , we worked from the raw detection data; for Sarell (2012) , we used the summary data reported in their own study. We then compiled all three datasets into a single dataframe recording the number of species identified per night at each site, along with site location and deployment and retrieval dates. Locations were kept separate even when they shared the same or similar names, so that distinct sites were not conflated in the summaries.

3.2.1 Data Submission

The long term acoustic monitoring data set for the Paterson (2024) and Skurikhina and Kellner (2025) studies has been formatted and submitted to BC Wildlife Species Inventory under project 6430 - South Okanagan Bat Hibernacula Project. Data has been also been submitted to NABat and can be viewed in the Partner Portal) under project: South Okanagan Bat Winter Bat Project (Project ID:16855). Analyzed acoustic files have been submitted to WildTrax and are available can be found under the project: South Okanagan Bat Winter Bat. Access to data sources will require permission from Mandy Kellner, and can be requested directly through the portals.

4 Results

Acoustic sampling happened in a total of 7 years, sampling a total of 20 unique sites Figure 1 . Across all survey years, Cougar Canyon had the highest number of recordings in a single year (2021), with a total of 13,428 detections. The greatest sampling effort in terms of site coverage occurred in 2011-12, during which 10 unique sites were monitored.

In total there have been 107978 bat recordings recorded over the length of the study. A total of 15 bat species were confirmed across all sites and years. The most frequently recorded species was LANO, with 20,051 detections.

To view summary species counts for each site refer to Appendix A

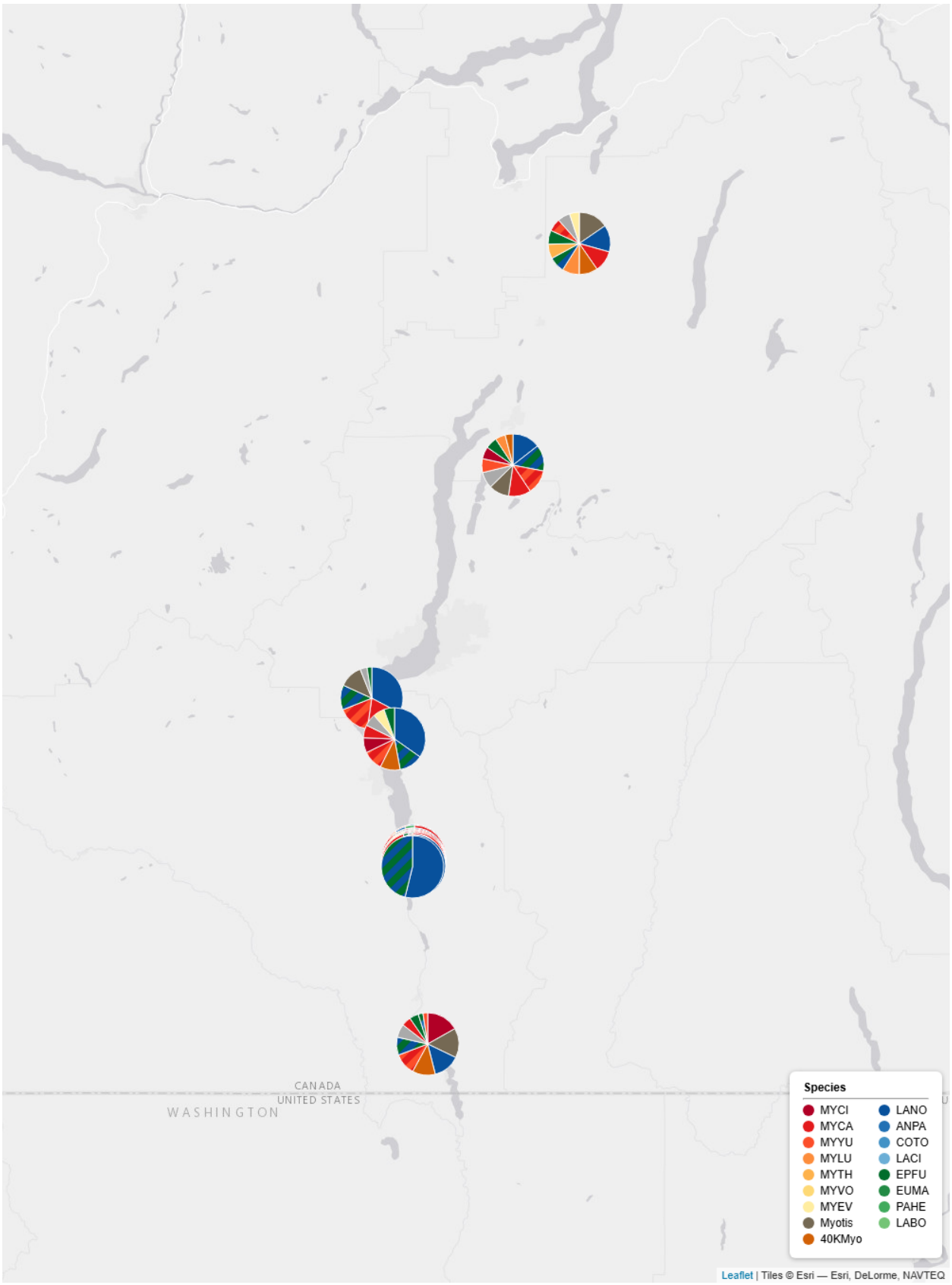


Figure 1: Proportion of species bat passes at each site. Only Species or species couplets that identified at least 5% of the total tags appear on map

5 Discussion

There are several underground cave/talus-cave systems known to BC Parks. Assessment of bat use, species presence, and timing of use could be accomplished through acoustic monitoring and/or guano collection. One such area is at Doctor's wall; during this survey, nets were set up near the entrances of identified caves, but no bats were detected during emergence or caught in the nets. One of the project partners set up an experimental acoustic detector inside the larger cave on November 12, 2024. As of March 2025, this detector had not been retrieved. Options for future work include additional placement of bat detectors at openings or inside caves, single-species DNA analysis through collection of single pellets of guano in paper envelopes (Wildlife Genetics International), and/or multi-species DNA analysis through collection of multiple guano samples (Northern Arizona University Species from Feces lab).

5.1 Recommendations

1. Explore use of detector array to increase understanding of bat winter movements and potential travel corridors as well as inform capture efforts.
2. Target Gilles Creek and the Spring for additional monitoring and capture work as the area currently believed to offer the best chance of successful captures.
3. Increase understanding of bat use of cave/talus features that are known or suspected to be used by bats, through passive acoustic monitoring and guano collection to determine timing of use and species.
4. Expand collection of existing data. This project aims to identify and include additional acoustic datasets. Possible sources including data from 2025/26 Skaha Park winter monitoring (Kellner pers comm), and ECCC/CWS year-round monitoring at Vaseaux (T. Luszcz pers comm)

6 Acknowledgements

This work was supported by funding from the BC Parks License Plate Program, as well as in-kind staff time, vehicles, and equipment from BC Parks and WLRS.

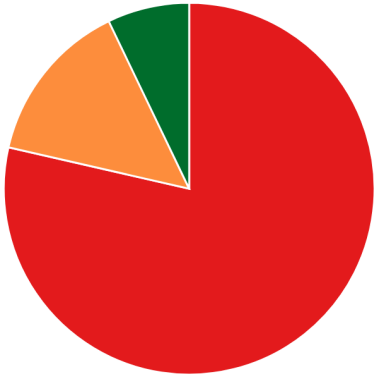
Data included in this compilation and summary was from work done for BC Parks by Brian Paterson and BC Water, Land and Resource Stewardship. Reports and summarized data are available on the BC Wildlife Species Inventory website.

7 Appendix A

Species detection summaries for each monitoring site.

Blazing

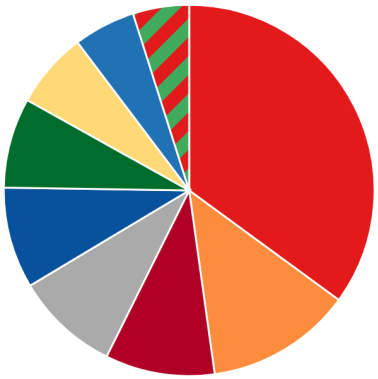
Coordinates: 49.44030, -119.56313
Source: Sarrell 2012



Species	n	%
EPFU	1	7.1%
MYCA	11	78.6%
MYLU	2	14.3%

Cave Hill

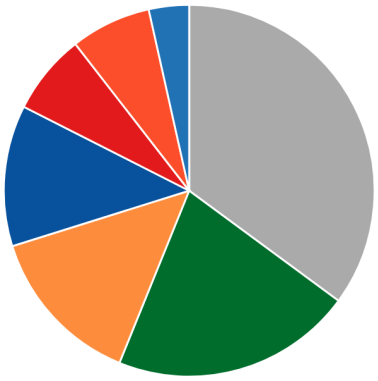
Coordinates: 49.44906, -119.56121
Source: Sarrell 2012



Species	n	%
ANPA	22	5.4%
COTO	8	2%
EPFU	32	7.8%
EUMA	1	0.2%
LANO	36	8.8%
MYCA	143	35%
MYCI	39	9.6%
MYEV	5	1.2%
MYLU	52	12.7%
MYVO	27	6.6%
MYYU	7	1.7%
HiF	1	0.2%
LoF	15	3.7%
MYCAPAHE	20	4.9%

Fern Gully

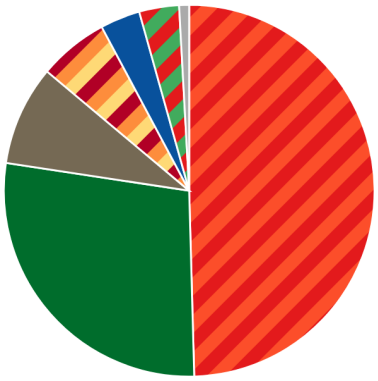
Coordinates: 49.45010, -119.56168
Source: Sarrell 2012



Species	n	%
ANPA	2	3.5%
COTO	1	1.8%
EPFU	12	21.1%
LANO	7	12.3%
MYCA	4	7%
MYLU	8	14%
MYYU	4	7%
LoF	19	33.3%

Fortress

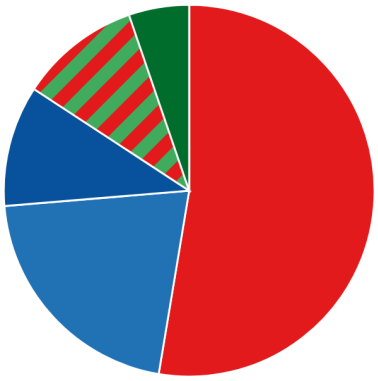
Coordinates: 49.43707, -119.56755
Source: Sarrell 2012



Species	n	%
ANPA	1	0.9%
EPFU	32	27.8%
LANO	4	3.5%
MYCAMYYU	57	49.6%
MYCAPAHE	4	3.5%
MYCIMYLUMYVO	7	6.1%
Myotis	10	8.7%

Go Anywhere

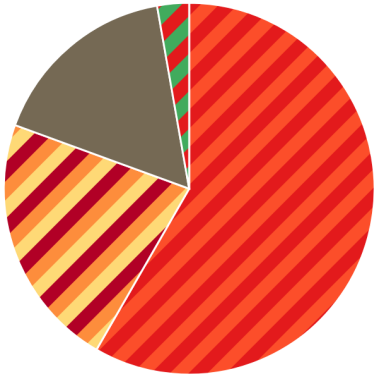
Coordinates: 49.44344, -119.56202
Source: Sarrell 2012



Species	n	%
ANPA	4	21.1%
EPFU	1	5.3%
LANO	2	10.5%
MYCA	10	52.6%
MYCAPAHE	2	10.5%

Great White

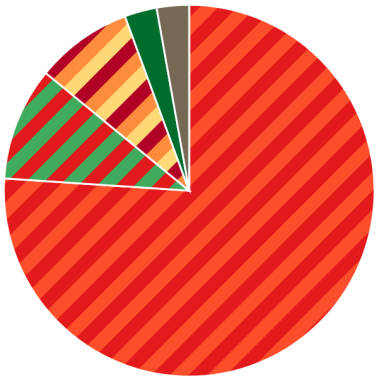
Coordinates: 49.44152, -119.56199
Source: Sarrell 2012



Species	n	%
MYCAMYYU	21	58.3%
MYCAPAHE	1	2.8%
MYCIMYLUMYVO	8	22.2%
Myotis	6	16.7%

No Climbs

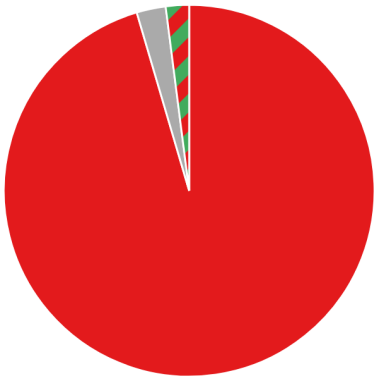
Coordinates: 49.43901, -119.56274
Source: Sarrell 2012



Species	n	%
EPFU	2	2.8%
MYCAMYYU	54	76.1%
MYCAPAHE	7	9.9%
MYCIMYLUMYVO	6	8.5%
Myotis	2	2.8%

Oph6B-Unknown

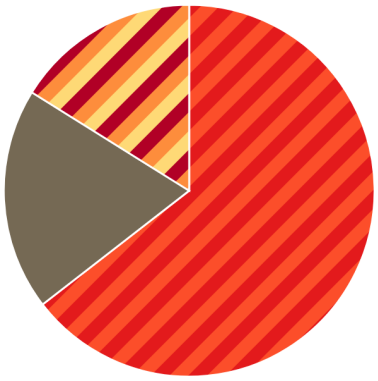
Coordinates: 49.43568, -119.56273
Source: Sarrell 2012



Species	n	%
EPFU	2	1%
MYCA	188	95.4%
MYLU	3	1.5%
MYCAPAHE	4	2%

Wave

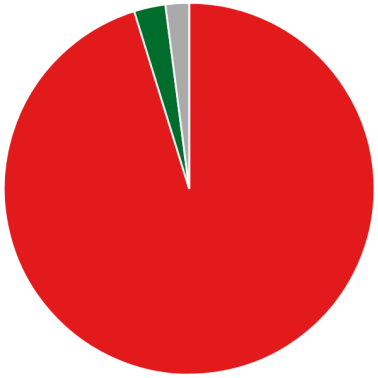
Coordinates: 49.43848, -119.56528
Source: Sarrell 2012



Species	n	%
MYCAMYYU	20	64.5%
MYCIMYLUMYVO	5	16.1%
Myotis	6	19.4%

Whitewash

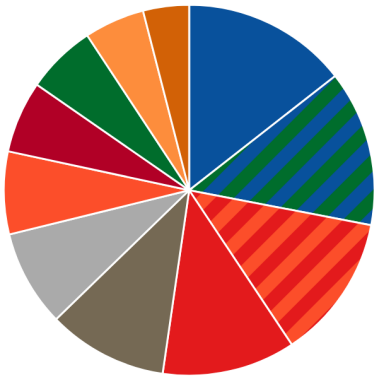
Coordinates: 49.43475, -119.56659
Source: Sarrell 2012



Species	n	%
EPFU	12	2.7%
LANO	8	1.8%
MYCA	417	95.2%
MYCAPAHE	1	0.2%

Cougar Canyon

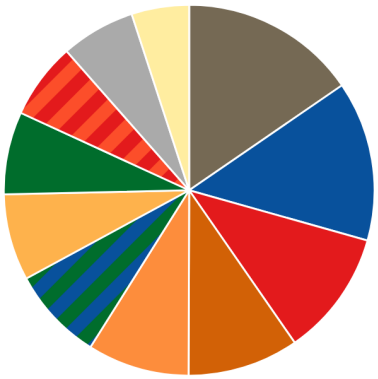
Coordinates: 50.18060, -119.27510
Source: Paterson 2024



Species	n	%
ANPA	1	0%
COTO	512	1.6%
EPFU	1911	6.1%
LABO	15	0%
LACI	8	0%
LANO	4534	14.4%
MYCA	3654	11.6%
MYCI	1969	6.3%
MYEV	121	0.4%
MYLU	1654	5.3%
MYTH	626	2%
MYVO	19	0.1%
MYYU	2265	7.2%
PAHE	578	1.8%
ANPAEPFU	21	0.1%
COTOEPFU	7	0%
COTOMYTH	15	0%
EPFULACI	1	0%
EPFULANO	4268	13.6%
EPFUMYEV	24	0.1%
EPFUMYTH	23	0.1%
LABOMYLU	3	0%
LoF	6	0%
MYCAMYCI	67	0.2%
MYCAMYYU	3970	12.6%
MYCAPAHE	1	0%
MYCIMYLU	27	0.1%
MYLUMYVO	7	0%
Myotis	3286	10.5%
MYYUPAHE	1	0%
20K	7	0%
25K	2	0%
30K	1	0%
40K	121	0.4%
40KMyo	1259	4%
45K	301	1%
50K	128	0.4%

Enderby Cliffs

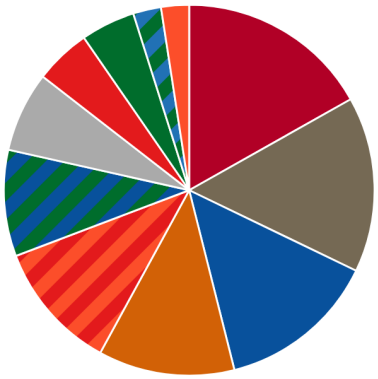
Coordinates: 50.59000, -119.08330
Source: Paterson 2024



Species	n	%
COTO	28	0.4%
EPFU	472	7.2%
EUMA	16	0.2%
LABO	3	0%
LACI	43	0.7%
LANO	913	13.9%
MYCA	719	11%
MYCI	29	0.4%
MYEV	329	5%
MYLU	582	8.9%
MYTH	494	7.5%
MYVO	69	1.1%
MYYU	85	1.3%
ANPAEPFU	2	0%
COTOEPFU	32	0.5%
COTOMYTH	1	0%
EPFULACI	1	0%
EPFULANO	534	8.2%
EPFUMYEV	70	1.1%
EPFUMYTH	12	0.2%
LABOMYCA	2	0%
LABOMYLU	3	0%
LoF	25	0.4%
MYCAMYCI	1	0%
MYCAMYYU	435	6.6%
MYCAPAHE	1	0%
Myotis	1010	15.4%
25K	1	0%
40KMyo	635	9.7%

Haynes Lease

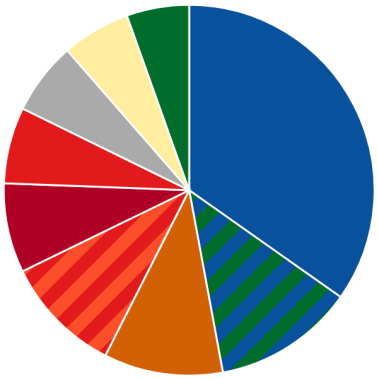
Coordinates: 49.09400, -119.52260
Source: Paterson 2024



Species	n	%
ANPA	371	1.5%
COTO	136	0.5%
EPFU	1184	4.7%
EUMA	230	0.9%
LABO	3	0%
LACI	6	0%
LANO	3476	13.9%
MYCA	1219	4.9%
MYCI	4206	16.8%
MYEV	16	0.1%
MYLU	492	2%
MYTH	95	0.4%
MYVO	50	0.2%
MYYU	609	2.4%
PAHE	30	0.1%
ANPACOTO	5	0%
ANPAEPFU	609	2.4%
COTOEPFU	13	0.1%
COTOMYTH	3	0%
EPFULACI	2	0%
EPFULANO	2317	9.3%
EPFUMYEV	7	0%
EPFUMYTH	7	0%
LABOMYCA	1	0%
LABOMYLU	7	0%
LABOMyo	5	0%
LoF	10	0%
MYCAMYCI	77	0.3%
MYCAMYYU	2813	11.3%
MYCAPAHE	4	0%
MYCIMYLU	25	0.1%
Myotis	3825	15.3%
MYYUPAHE	8	0%
25K	1	0%
30K	51	0.2%
40K	21	0.1%
40KMyo	2972	11.9%
45K	64	0.3%
50K	7	0%

Okanagan Mountain

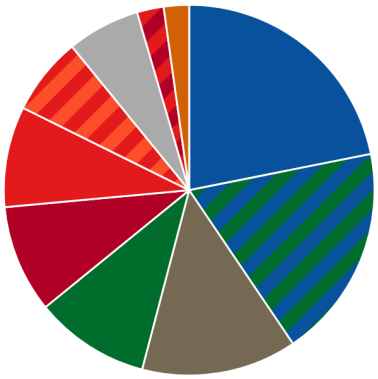
Coordinates: 49.67000, -119.61820
Source: Paterson 2024



Species	n	%
ANPA	1	0%
COTO	78	1%
EPFU	414	5.5%
EUMA	43	0.6%
LABO	2	0%
LACI	2	0%
LANO	2641	34.8%
MYCA	506	6.7%
MYCI	592	7.8%
MYEV	453	6%
MYLU	66	0.9%
MYTH	119	1.6%
MYVO	15	0.2%
MYYU	31	0.4%
PAHE	3	0%
ANPAEPFU	8	0.1%
COTOEPFU	2	0%
COTOMYTH	1	0%
EPFULANO	935	12.3%
EPFUMYEV	35	0.5%
EPFUMYTH	2	0%
LACITABR	1	0%
LoF	4	0.1%
MYCAMYCI	2	0%
MYCAMYYU	784	10.3%
MYCIMYLU	5	0.1%
MYEVMYTH	1	0%
Myotis	54	0.7%
25K	2	0%
30K	4	0.1%
40KMyo	789	10.4%

Skaha - Fern Gully

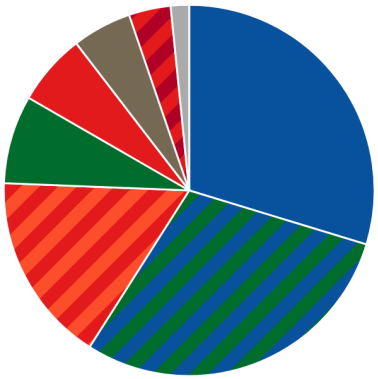
Coordinates: 49.44340, -119.56560
Source: Paterson 2024



Species	n	%
ANPA	3	0%
COTO	464	1.6%
EPFU	2996	10%
EUMA	8	0%
LABO	3	0%
LACI	12	0%
LANO	6527	21.9%
MYCA	2622	8.8%
MYCI	2809	9.4%
MYEV	35	0.1%
MYLU	176	0.6%
MYTH	272	0.9%
MYVO	10	0%
MYYU	581	1.9%
PAHE	3	0%
COTOEPFU	36	0.1%
COTOMYTH	10	0%
EPFULACI	5	0%
EPFULANO	5574	18.7%
EPFUMYEV	10	0%
EPFUMYTH	5	0%
LoF	8	0%
MYCAMYCI	695	2.3%
MYCAMYYU	2025	6.8%
MYCAPAHE	4	0%
MYCIMYLU	128	0.4%
MYLUMYVO	9	0%
Myotis	4046	13.6%
25K	5	0%
30K	28	0.1%
40KMyo	659	2.2%
45K	49	0.2%
50K	35	0.1%

Skaha - Great White

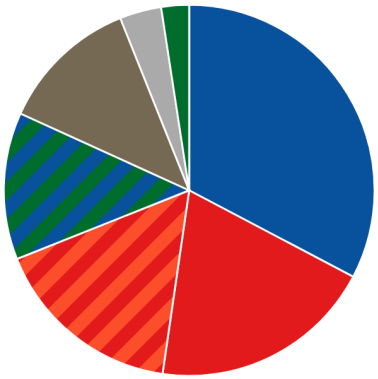
Coordinates: 49.44240, -119.56190
Source: Paterson 2024



Species	n	%
COTO	6	0.5%
EPFU	101	7.7%
LANO	390	29.7%
MYCA	82	6.2%
MYEV	1	0.1%
MYTH	2	0.2%
MYYU	2	0.2%
EPFULANO	385	29.3%
EPFUMYEV	1	0.1%
MYCAMYCI	48	3.7%
MYCAMYYU	218	16.6%
Myotis	68	5.2%
40KMyo	7	0.5%
45K	2	0.2%

Wildhorse Canyon

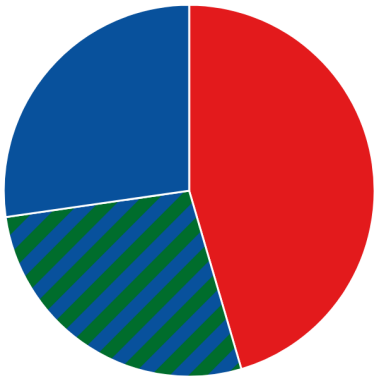
Coordinates: 49.74500, -119.68380
Source: Paterson 2024



Species	n	%
COTO	3	0.1%
EPFU	114	2.4%
EUMA	80	1.7%
LACI	1	0%
LANO	1538	32.7%
MYCA	921	19.6%
MYCI	16	0.3%
MYEV	1	0%
MYTH	10	0.2%
MYYU	20	0.4%
COTOEPFU	1	0%
EPFULANO	603	12.8%
EPFUMYEV	3	0.1%
EPFUMYTH	4	0.1%
LoF	1	0%
MYCAMYCI	2	0%
MYCAMYYU	784	16.7%
MYCAPAHE	1	0%
MYCIMYLU	3	0.1%
Myotis	571	12.1%
25K	1	0%
40KMyo	24	0.5%

Gillies Creek Open Forest Detector

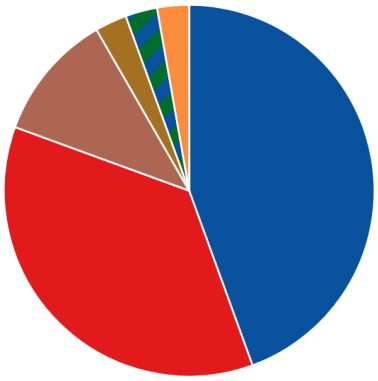
Coordinates: 49.43122, -119.56193
Source: Skurikhina and Kellner 2025



Species	n	%
LANO	3	27.3%
MYCA	5	45.5%
EPFULANO	3	27.3%

Gillies Creek Road Detector

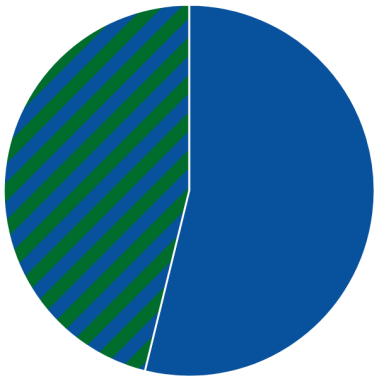
Coordinates: 49.43010, -119.56220
Source: Skurikhina and Kellner 2025



Species	n	%
LANO	16	44.4%
MYCA	13	36.1%
MYLU	1	2.8%
EPFULANO	1	2.8%
40K	1	2.8%
50K	4	11.1%

Smythe Drive Spring Detector

Coordinates: 49.42836, -119.56680
Source: Skurikhina and Kellner 2025



Species	n	%
LANO	7	53.8%
EPFULANO	6	46.2%

Literature Cited

Blejwas, Karen, Grey Pendleton, Michael Kohan, and Laura Beard. 2021. "The Milieu Souterrain Superficiel as Hibernation Habitat for Bats: Implications for White-Nose Syndrome." *Journal of Mammalogy* 102 (June): 1110–27. https://doi.org/10.1093/jmammal/gyab050.

Lausen, Cori, and Jason Rae. 2018. *BatCaver in British Columbia. Data Summary and Recommendations for Effective Conservation Action*. https://a100.gov.bc.ca/pub/siwe/download.do?docId=33381.

Nagorsen, David, Andrew Bryant, David Kerridge, Gina Roberts, Anna Roberts, and Michael Sarell. 1994. "Winter Bat Records for British Columbia." *Northwestern Naturalist* 74 (December): 61. https://doi.org/10.2307/3536599.

Paterson, Brian. 2024. *Okanagan Winter Bat Monitoring: 5 Year Summary 2018-2023.*

Sarell, Mike. 2012. *Late Winter Activity of Bats at Skaha Bluffs*.

Skurikhina, Anna, and Mandy Kellner. 2025. *South Okanagan Bat Hibernacula Microsite Pilot Project. 2024-25 Field Update.*

Weller, Theodore J., Thomas J. Rodhouse, Daniel J. Neubaum, et al. 2018. "A Review of Bat Hibernacula Across the Western United States: Implications for White-Nose Syndrome Surveillance and Management." *PLOS ONE* 13 (10): e0205647. https://doi.org/10.1371/journal.pone.0205647.